Sessions 4-5 - Managing transaction exposure

Understand how FX quotations, forwards and options change a firm's cash flows, and calculate hedge outcomes without mixing currencies, dates or exposure amounts.

Plan

1. Identify the risk and the currency exposure
2. Read exchange rates and forward prices
3. Match the hedge to a receivable or a payable
4. Aidan: a GBP receivable in EUR
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6. Over-hedging: separate currency risk from quantity risk
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1. Identify the risk and the currency exposure

- A firm faces several connected risks: market, credit, liquidity, operational, foreign exchange, interest rate, business, reputational and commodity risk.
- Risk management starts by identifying what can change, which cash flow it affects, and when the firm must pay or collect it.
- The A2A example distinguishes liquidity risk, derating risk, interest rate risk and sustainability covenant risk. Its responses include financial planning and pre-funding, asset rotation and equity/hybrid financing, fixed/variable-rate debt management, and a sustainable financial committee.

> **Transaction exposure:** gains or losses from settling existing financial obligations whose terms are stated in a foreign currency. It concerns contractual cash flows, not simply a forecast of future sales.

- Examples: a sale invoiced in GBP by a EUR-based firm; a USD-based firm's GBP supplier payable; foreign-currency loan repayment; assets or liabilities denominated in a foreign currency.
- **Operating/economic exposure:** expected future cash flows may change because exchange-rate movements affect international competitiveness.
- **Translation/accounting exposure:** FX volatility affects consolidation of subsidiaries with different currencies.
- Do not treat these as interchangeable. This lecture's contractual derivatives primarily manage **transaction exposure**.

> **Receivable:** foreign currency the firm will receive. The firm is hurt if that currency loses value in home currency.
> **Payable:** foreign currency the firm must pay. The firm is hurt if that currency becomes more expensive in home currency.

- Hedging reduces the variability of expected cash flows. It is not a promise that the hedged outcome will beat the unhedged outcome in every scenario.
- A treasury decision depends on risk tolerance, the firm's exchange-rate view and its confidence in that view. Removing downside also has a cost: a premium or forgone favourable movements.

2. Read exchange rates and forward prices

- **Spot market:** almost immediate purchase or sale of foreign exchange.
- **Forward market:** agree today to buy or sell foreign exchange on a future date at a fixed price. In the lecture's setup, no cash changes hands at entry; exchange occurs at maturity.
- Write the units before calculating. In $S_{j/k}$, the quotation is currency j per one unit of currency k.
- For a European investor, **direct quotation** is EUR/USD: euros per dollar. **Indirect quotation** is USD/EUR: dollars per euro.

Direct rate = 1 / Indirect rate
EUR proceeds = USD amount / (USD/EUR rate) = USD amount × (EUR/USD rate)

Where:

- The USD amount is the foreign-currency cash flow.
- The indirect rate is dollars per euro; its reciprocal is euros per dollar.

What this does:

- First check the denominator of the quote. If one euro buys 1.60 dollars, one dollar buys 1/1.60 = EUR 0.625.
- Then convert the cash flow: USD 100 / 1.60 = EUR 62.50.
- If USD/EUR rises, the euro strengthens and a fixed USD receivable yields **fewer euros**. If EUR/USD rises, the dollar strengthens and it yields **more euros**.

2.1 Forward premium or discount

$$f_{N,j} = [(F_N - S) / S] \times (360 / N)$$

Where:

- S is today's spot quote, in USD per unit of currency j (American terms).
- F_N is the N-day forward quote, using exactly the same units.
- N is the maturity in days; the lecture uses a 360-day year.
- f is an annualised decimal. Multiply by 100 to express it as a percentage.

What this does:

- Subtract spot from forward to find the price difference.
- Divide by spot to make it a relative difference.
- Multiply by 360/N to annualise. Positive means a premium for j in this quotation; negative means a discount.

EUR example from Exhibit 5.7: S = USD 1.1235/EUR and six-month F = USD 1.1299/EUR. Using N = 180 days: [(1.1299 - 1.1235)/1.1235] × 2 = 0.011393, or **1.1393% annualised premium**.

- Do not reverse the quote halfway through. Reversing a quotation changes which currency's premium/discount you are describing.
- Keep enough decimals before rounding: the yen quotes in the displayed table are rounded, so they do not reproduce the slide's 0.460% answer exactly.

3. Match the hedge to a receivable or a payable

Use a **direct quote** throughout this section: S_T, F and K are home currency per unit of foreign currency. Q is the foreign-currency amount; p_T is the premium per foreign unit, expressed in home currency at the comparison date T.

Cash flow	No hedge	Forward	Bought option
Receive foreign currency	Sell Q at spot at T	Short forward: sell Q at F	Put: right to sell Q at K
Pay foreign currency	Buy Q at spot at T	Long forward: buy Q at F	Call: right to buy Q at K

Receivable: unhedged proceeds = $Q \times S_T$
Forward proceeds = $Q \times F$
Put-hedged net proceeds = $Q \times [\max(S_T, K) - p_T]$

What this does:

- The exporter sells the foreign currency. Choose the **higher selling price**, spot or strike.
- Exercise the put when $S_T < K$; otherwise sell at spot. At equality, either gives the same gross amount.
- Deduct the premium in both branches. The minimum net proceeds are $Q \times (K - p_T)$.

Payable: unhedged cost = $Q \times S_T$
Forward cost = $Q \times F$
Call-hedged total cost = $Q \times [\min(S_T, K) + p_T]$

What this does:

- The importer buys the foreign currency. Choose the **lower buying price**, spot or strike.
- Exercise the call when $S_T > K$; otherwise buy at spot.
- Add the premium in both branches. The maximum total cost is $Q \times (K + p_T)$.

3.1 Compare cash flows at the same date

- A premium is paid today, but the receivable/payable settles later. If the example accounts for financing/opportunity cost, convert the premium to its future value before comparing maturity cash flows.
- With a simple annual rate r and horizon t years, $p_T = p_0 \times (1 + r \times t)$. Here p_0 is today's premium per unit in home currency.
- First multiply the premium by the stated time factor; then subtract it from proceeds or add it to costs. Do not change the strike when doing this.
- **Exercise threshold** depends on strike K, not on premium. **Forward-option break-even** compares net cash flows and therefore includes the premium.

3.2 Understand the forward payoff, not just the contract

- A long forward has payoff $Q \times (S_T - F)$; a short forward has payoff $Q \times (F - S_T)$. These are gains/losses relative to the maturity spot price, in home currency.
- First find the spot-forward difference, then multiply by Q. A forward's standalone loss can be offset by a favourable change in the underlying commercial cash flow.
- In the GBP 100m importer example, F = USD 1.50/GBP. At $S_T = 1.80$, the unhedged purchase costs USD 180m and the long forward gains USD 30m. Net cost = USD 150m.
- At $S_T = 1.20$, the purchase costs USD 120m but the forward loses USD 30m. Net cost is still USD 150m. The hedge removes FX variation, not the purchase itself.

4. Aidan: a GBP receivable in EUR

- Aidan is a European exporter receiving **GBP 1,000,000 in 90 days** for equipment sold to a British customer.
- Use the slide and workbook inputs: spot EUR 1.09/GBP; forward EUR 1.076/GBP; put strike EUR 1.07/GBP; premium 1.5%; annual cost of capital 12%.
- A weaker pound reduces euro proceeds. The correct forward is **short GBP**; the correct option is a **bought GBP put**.

4.1 Forward hedge

- Sell GBP 1,000,000 forward at EUR 1.076/GBP.
- Maturity proceeds = 1,000,000 × 1.076 = **EUR 1,076,000**, whatever the maturity spot rate.
- This fixes the euro receipt, but gives up the benefit of a stronger pound.

4.2 Put option hedge

- Premium today = 1,000,000 × 1.5% × 1.09 = **EUR 16,350**.
- The percentage premium is applied to the notional converted at today's spot, not directly subtracted as EUR 0.015/GBP.
- Three-month future-value premium = 16,350 × (1 + 12%/4) = **EUR 16,840.50**.
- Per-pound maturity premium = 16,840.50 / 1,000,000 = **EUR 0.0168405/GBP**.

Net proceeds = 1,000,000 × max(S_T, 1.07) - 16,840.50
Floor = EUR 1,053,159.50
Forward-option break-even S* = 1.076 + 0.0168405 = EUR 1.0928405/GBP

What this does:

- First choose spot or strike, whichever yields more euros.
- Multiply by the GBP receivable and deduct the premium at maturity.
- To find break-even above the strike, set 1,000,000 × S **- 16,840.50 equal to EUR 1,076,000. Solve for S**.

S_T (EUR/GBP)	Unhedged (EUR)	Forward (EUR)	Put net (EUR)
1.00	1,000,000	1,076,000	1,053,159.50
1.07	1,070,000	1,076,000	1,053,159.50
1.10	1,100,000	1,076,000	1,083,159.50
1.20	1,200,000	1,076,000	1,183,159.50

- The put is exercised below **1.07**, not below the forecast rate 1.085. The forecast is a view, not a contractual exercise threshold.
- Below 1.0928405, the forward gives higher proceeds; above it, the put gives higher proceeds. This comparison assumes the stated premium convention and same notional/date.

4.3 The money-market line in the source chart

- The source chart also shows a money-market alternative. Borrow GBP now against the future receivable, convert to EUR today, then carry the euros to maturity.
- Using the slide's GBP borrowing rate of 10% annually: GBP borrowed = 1,000,000 / 1.025 = GBP 975,609.76. Conversion at 1.09 gives EUR 1,063,414.63 today.
- The chart's EUR 1,095,317 uses a **3% three-month opportunity-return assumption**: 1,063,414.63 × 1.03 = EUR 1,095,317.07.
- Investing instead at the stated EUR deposit rate of 6% annually gives 1,063,414.63 × 1.015 = EUR 1,079,365.85. The different totals reflect different euro carrying assumptions, not a different FX hedge.

5. Boeing: compare forwards and options in USD

5.1 Exporter: GBP 10m receivable in one year

- Boeing sells to British Airways and will receive **GBP 10,000,000**.
- Inputs: F = K = USD 1.46/GBP; put premium = USD 0.02/GBP.
- The slide's comparison deducts the stated premium directly; no financing adjustment is supplied for this example.

Forward receipt = 10,000,000 × 1.46 = USD 14,600,000
Put net receipt = 10,000,000 × [max(S_T, 1.46) - 0.02]
Put floor = USD 14,400,000
Break-even S* = 1.46 + 0.02 = USD 1.48/GBP

What this does:

- Below 1.46, exercise the put and receive USD 14.6m gross, then deduct USD 0.2m premium.
- Above 1.46, sell at spot and still deduct the premium.
- At spot 1.48, put net proceeds equal the forward's USD 14.6m. Above 1.48, the put yields more; below it, the forward yields more.

S_T (USD/GBP)	Forward receipt	Put net receipt
1.40	USD 14.60m	USD 14.40m
1.48	USD 14.60m	USD 14.60m
1.55	USD 14.60m	USD 15.30m

5.2 Importer: GBP 5m payable in one year

- This is a **different transaction**: Boeing imports an engine and owes **GBP 5,000,000**.
- Inputs: spot USD 1.80/GBP; forward USD 1.75/GBP; call strike USD 1.80/GBP; premium USD 0.018/GBP; US annual rate 6%.
- Buy GBP forward (**long**) or buy a GBP **call**. A put would protect the wrong direction.
- Upfront premium = 5,000,000 × 0.018 = **USD 90,000**.
- Future-value premium = 90,000 × 1.06 = **USD 95,400**, or USD 0.01908/GBP.

Forward cost = 5,000,000 × 1.75 = USD 8,750,000
Call total cost = 5,000,000 × min(S_T, 1.80) + 95,400
Maximum call cost = USD 9,095,400
Break-even S* = 1.75 - 0.01908 = USD 1.73092/GBP

What this does:

- First choose the cheaper way to buy GBP: spot or strike.
- Multiply by GBP 5m and add the maturity premium.
- Below the strike, set 5,000,000 × S **+ 95,400 equal to USD 8,750,000, then solve for S**.

S_T (USD/GBP)	Forward cost	Call total cost
1.65	USD 8,750,000	USD 8,345,400
1.73092	USD 8,750,000	USD 8,750,000
1.80 or above	USD 8,750,000	USD 9,095,400

- The call is cheaper than the forward below 1.73092; the forward is cheaper above it. The call is exercised only above 1.80.
- Premium paid today (90,000), premium at maturity (95,400), break-even (1.73092) and strike (1.80) are four different quantities.

6. Over-hedging: separate currency risk from quantity risk

> **Over-hedging:** the hedge notional exceeds the actual exposure. The excess derivative is no longer offset by the commercial cash flow and becomes a speculative position.

- At **T0, budget**, a European exporter expects USD 100 revenue, costs EUR 70 and an indirect budget rate of USD 1.30/EUR.
- The slide rounds the reciprocal to EUR 0.77/USD, giving a budget margin of EUR 7. Using the exact reciprocal gives 100/1.30 - 70 = **EUR 6.9231**.
- With revenue fixed at USD 100, a weaker dollar at USD 1.60/EUR gives a margin of 100/1.60 - 70 = **EUR -7.50**. A stronger dollar at USD 1.00/EUR gives **EUR 30**.
- These are currency effects **assuming the revenue amount remains USD 100**.

6.1 What happens when the amount also changes

- The CFO sells **USD 100 forward** at USD 1.30/EUR at T0.
- At **T1, invoicing**, actual sales are only **USD 50**. The remaining USD 50 forward commitment is now uncovered.
- At **T2, collection**, spot is USD 1.00/EUR. The firm collects USD 50, but still must deliver USD 100 under the forward.

Step at T2	EUR cash flow
Convert collected USD 50 at forward: 50 / 1.30	+38.4615
Buy missing USD 50 at spot: 50 / 1.00	-50.0000
Deliver these USD 50 at forward: 50 / 1.30	+38.4615
Net FX cash receipts	+26.9231
Operating costs	-70.0000
Final margin	**-43.0769**

- The excess hedge alone loses **EUR 11.5385**: EUR 38.4615 received minus EUR 50 paid to acquire the missing dollars.
- Actual sales converted at the forward yield EUR 38.4615. After the excess-hedge loss and costs, the final margin is EUR -43.0769, approximately the slide's EUR -43.
- The decline from exact budget margin EUR 6.9231 to EUR -43.0769 is **EUR 50**. It is **not all a derivative loss**.
- Decomposition: revenue shortfall at the locked rate = 50/1.30 = EUR 38.4615; extra hedge loss = EUR 11.5385; total deterioration = EUR 50.
- Against leaving the actual USD 50 unhedged at spot 1.00, the final margin would have been EUR -20. The hedged margin is worse by EUR 23.0769: forgone appreciation on collected dollars plus the uncovered-forward loss.

6.2 Design the hedge around the exposure

- Ask **when** to hedge: at budget, the cash flow is still uncertain; at invoicing, the contractual amount is clearer.
- Ask **how much** to hedge: hedging 100% of a forecast removes price uncertainty on that notional but does not remove quantity uncertainty.
- Keep reviewing the commercial cash flow and hedge amount together. A derivative can become speculative when the underlying sale disappears or shrinks.

7. Recap: budget, invoicing and collection

- A European exporter expects and invoices **USD 100**. Collection occurs at T2.
- Two T2 scenarios: **USD 1.40/EUR** (stronger dollar, favourable to the exporter) and **USD 1.75/EUR** (weaker dollar, unfavourable).
- Hedge choices: T0 forward at USD 1.54/EUR; T1 forward at USD 1.65/EUR; or a T0 put protecting sale of USD at the strike.
- Use the **worked solution's precise T0/strike quote, USD 1.5758/EUR**, consistently. Its reciprocal is **EUR 0.6345983/USD**. The earlier setup's 1.57 is a different, less precise input and must not be mixed with the more precise solution.

7.1 Premium and net option proceeds

- Premium is USD 0.02 **per dollar sold**, so total premium is USD 2.
- Convert that premium at T0: USD 2 / 1.5758 = **EUR 1.2692**. This exercise deducts that cost directly, without a financing adjustment.
- Direct strike k = 1/1.5758 = EUR 0.6345983/USD. Direct maturity spot $s_T = 1/R_T$, where R_T is USD/EUR.

Put net proceeds = $100 \times \max(s_T, k) - 1.2691966$
Option floor = 100/1.5758 - 2/1.5758 = EUR 62.1906

What this does:

- Convert the indirect quote to euros per dollar before comparing spot with strike.
- At $R_T = 1.75$, each dollar buys fewer euros than the strike: exercise the put.
- At $R_T = 1.40$, spot gives more euros: let the put expire and sell at spot.
- Deduct the same EUR 1.2692 premium in both cases.

Strategy: realised EUR revenue	**$R_T = 1.40$**	**$R_T = 1.75$**
No hedge: $100/R_T$	71.4286	57.1429
T0 forward: 100/1.54	64.9351	64.9351
T1 forward: 100/1.65	60.6061	60.6061
Put: best sale rate minus premium	70.1594	62.1906

- Net effective EUR/USD rates for the option: **0.701594** in the favourable scenario and **0.621906** at the floor.
- Their USD/EUR reciprocals are **1.425326** and **1.607959**. An effective rate is net proceeds divided by notional, not the market spot quote.
- Consistent budget gross revenue at 1.5758 is **EUR 63.4598**. The standalone EUR 62.5 budget label on the final slide is not the conversion of USD 100 at 1.5758.
- The forward produces a fixed receipt. The put has a lower guaranteed net floor than the T0 forward but preserves favourable FX movements after paying its premium.
- The T1 forward locks in less revenue because 1.65 dollars are now needed to buy one euro. It still protects against further adverse moves; a lower locked amount does not by itself make a hedge pointless.

8. Common mistakes & exam traps

- **Multiplying by an indirect quote:** USD to EUR uses USD ÷ (USD/EUR). Multiplication is correct only with EUR/USD.
- **Calling a higher USD/EUR quote a stronger dollar:** it is a stronger euro. The dollar receivable yields fewer euros.
- **Wrong hedge direction:** receivable = short forward or bought put; payable = long forward or bought call on the foreign currency.
- **Confusing strike with break-even:** exercise depends on strike; whether the option beats a forward depends on premium and forward rate too.
- **Subtracting premium from importer cost:** a bought option adds cost. For an exporter it reduces net proceeds.
- **Mixing today's and maturity money:** apply the stated future-value premium in Aidan and Boeing's importer case; use the direct premium convention of the Boeing exporter and recap examples.
- **Changing the example's currencies:** Aidan's outcomes are EUR; Boeing's are USD. A copied "euro/GBP" label does not change Boeing's USD cash flows.
- **Treating all budget deterioration as over-hedging loss:** separate missing revenue, lost favourable FX opportunity on actual revenue, and the uncovered derivative's loss.
- **Rounding inputs too early:** keep full precision in reciprocals and premium calculations; round final cash amounts. Approximate slide labels are not new inputs.
- **Reading a derivative payoff as the total business cash flow:** the hedge gain/loss offsets the underlying exposure; the invoice remains payable or receivable.

9. Must know absolutely

Core logic

- Transaction exposure concerns existing foreign-currency contractual obligations.
- Always state quotation units, foreign notional and payment date first.
- Exporter: maximise the selling rate; importer: minimise the buying rate.
- A forward fixes the rate and creates an obligation. A bought option preserves choice but costs a premium.
- Receivable put floor = $Q \times (K - p_T)$; payable call ceiling = $Q \times (K + p_T)$, with direct quotes.
- Aidan break-even: EUR 1.0928405/GBP. Boeing exporter: USD 1.48/GBP. Boeing importer: USD 1.73092/GBP.
- A hedge sized to forecast revenue can become speculative if actual revenue is lower.
- Compare complete net cash flows in the same home currency at the same date.

Source basis. Rachele Anconetani, **Corporate Financial Risk Management: Managing Transaction Exposure**, Financial Management & Markets, 2026/2027, slides 4-43; accompanying Aidan and Boeing workbooks. Calculations reworked from the stated inputs. Revised 13 October 2026.